

Extending Active Recommendation for Email Outreach Dynamics

Refinements to a SAE + Thompson Sampling Bandit

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1 Overview

- **Problem:** cold-start email campaigns — new templates have no interaction history
- **Baseline:** SAE-based collaborative filtering + Thompson Sampling bandit over recipients
- **This document:** recaps the baseline, presents 9 refinements, and compares results

2 Baseline Model

Interaction matrix $X \in \{0, 1\}^{n \times m}$, $X_{i,j} = 1$ iff recipient j opened template i .

$$\min_{E,D} \ell(X, \sigma(XB_{E,D})), \quad B_{E,D} = ED^\top - \text{diag}([E \odot D]\mathbf{1})$$

$$\Sigma = \sigma(B_{E,D}) \Rightarrow \Sigma_{i,j} = P(j \text{ opens} \mid i \text{ already opened})$$

Each recipient j is scored as:

$$s_j(t) = \alpha \cdot \phi_j p_j + (1 - \alpha) \cdot f_j(t)$$

- ϕ_j — historic open rate of recipient j
- p_j — influence of j on other recipients
- $f_j(t)$ — confidence j opens, given current state

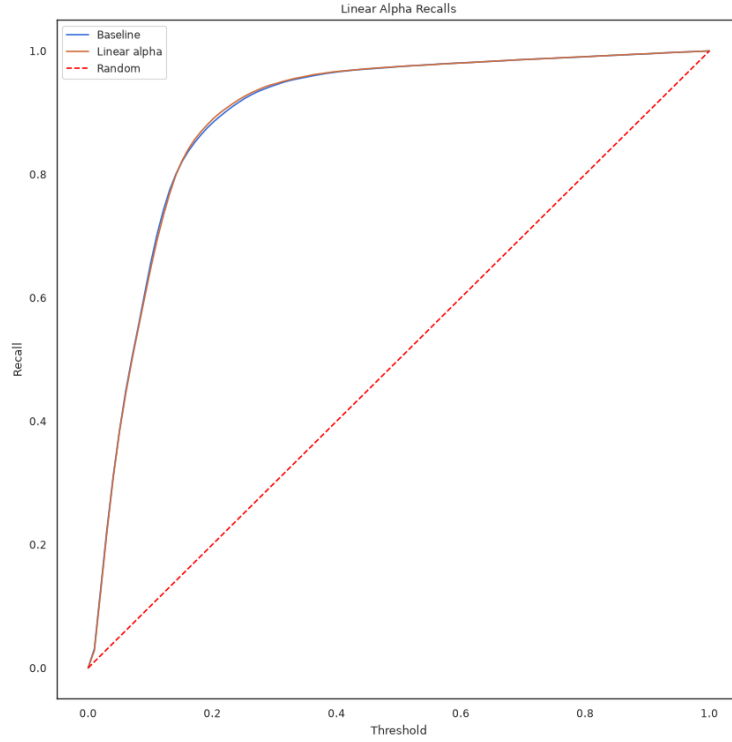
$$\alpha_j(t) = G s_j(t), \quad \beta_j(t) = G(1 - s_j(t))$$

3 Linear Alpha Scheduling

Problem: a fixed α cannot shift from exploration to exploitation as the campaign progresses.

$$\alpha(\mu) = l(1 - \mu) + r\mu, \quad \mu = \frac{N(t)}{N}$$

Result ($l = 0.1, r = 0.05$): R@5/15/25/35% = 0.383 / 0.823 / 0.927 / 0.960 AUC = 0.894

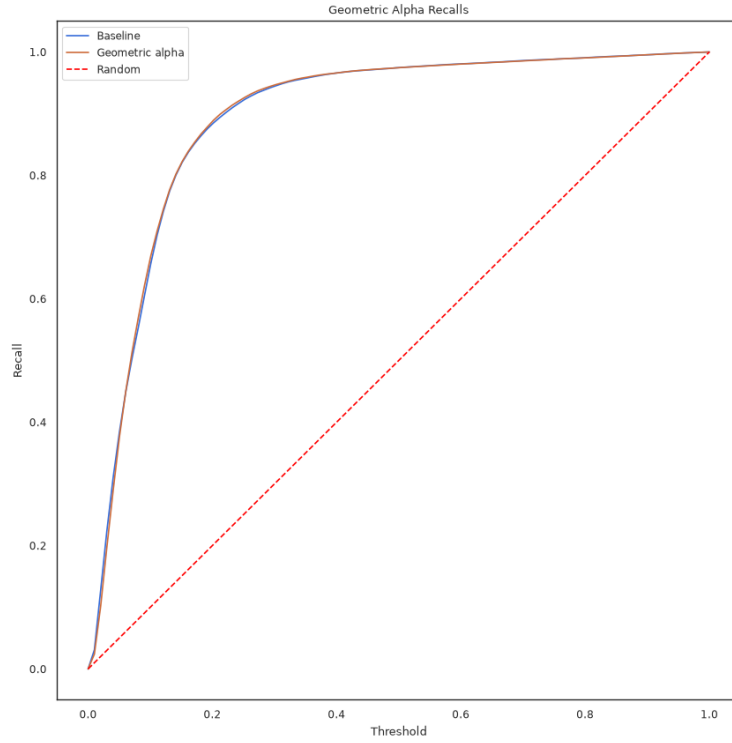


4 Geometric Alpha Scheduling

Problem: same motivation as linear scheduling; a log-linear decay may fit the explore/exploit shift better.

$$\alpha(\mu) = l^{(1-\mu)} r^\mu$$

Result ($l = 0.3$, $r = 0.05$): R@5/15/25/35% = 0.377 / 0.823 / 0.927 / 0.959 AUC = 0.894

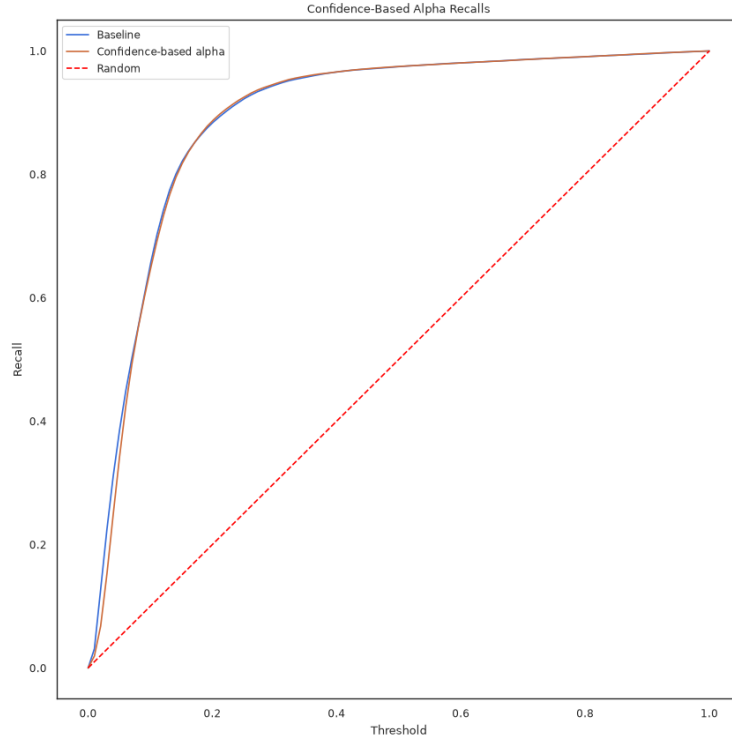


5 Confidence-Based Alpha Scheduling

Problem: trust in $f_j(t)$ should grow as more opens are observed, not just as time passes.

$$\alpha(\tilde{o}) = \frac{\kappa}{\tilde{o} + \kappa}, \quad \tilde{o} = \frac{o(t)}{m}$$

Result ($\kappa = 0.003$): R@5/15/25/35% = 0.341 / 0.818 / 0.926 / 0.960
AUC = 0.891

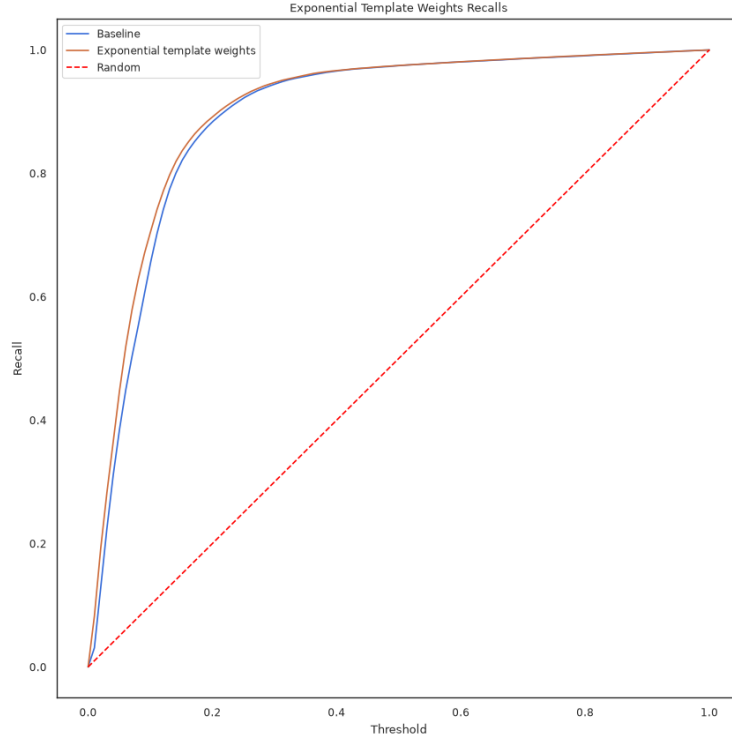


6 Recency-Weighted Historical Engagement

Problem: ϕ_j weights all past templates equally, ignoring behavioral drift over time.

$$\omega_i = 2^{-d_i/h}, \quad \phi_j = \frac{\omega^\top X_{:,j}}{\sum_i \omega_i}$$

Result ($h = 0.3$): R@5/15/25/35% = 0.447 / 0.837 / 0.928 / 0.960 AUC = 0.903

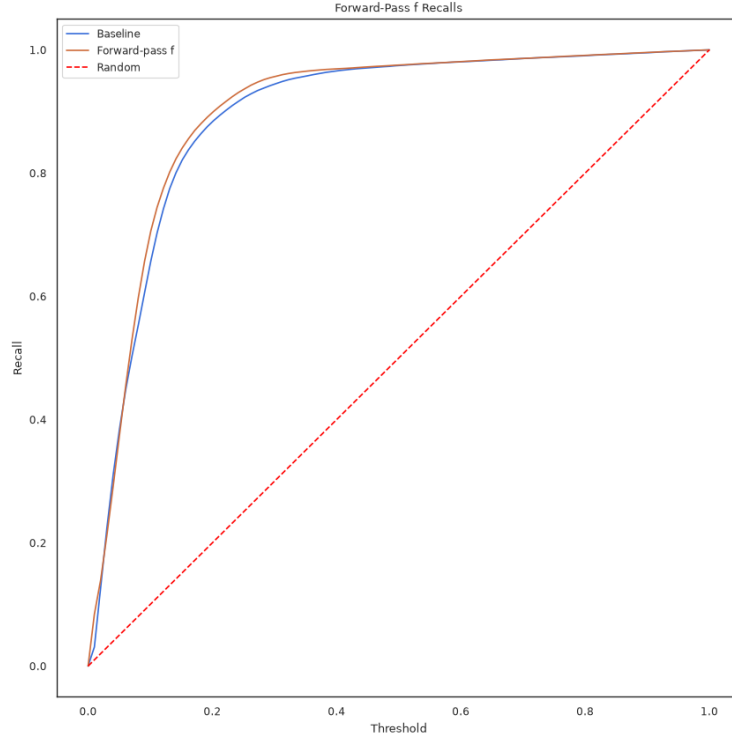


7 Forward-Pass Definition of $f_j(t)$

Problem: the linear form of $f_j(t)$ cannot capture joint influence of multiple simultaneous openers.

$$f_j(t) = f_{SAE}(x(t)) e_j = \sigma(x(t)^\top B_{E,D}) e_j$$

Result: R@5/15/25/35% = 0.375 / 0.841 / 0.937 / 0.966 AUC = 0.901

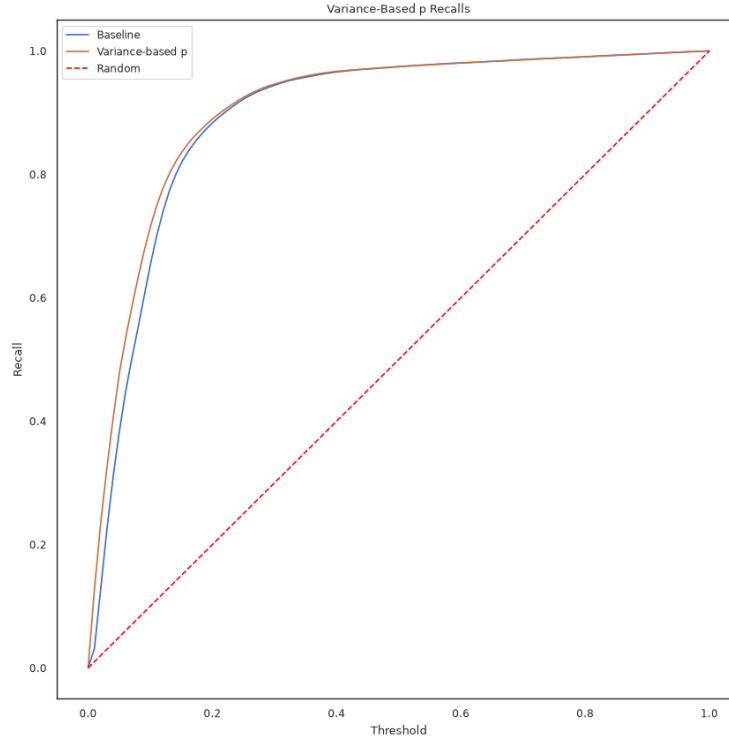


8 Variance-Based Definition of p_j

Problem: baseline p_j rewards score *increase*, not the *uncertainty reduction* that defines information gain.

$$p_j = \frac{4}{m-1} \sum_{i \neq j} (\Sigma_{j,i} - \bar{\Sigma}_{j,:})^2$$

Result: R@5/15/25/35% = 0.479 / 0.837 / 0.926 / 0.960 AUC = 0.905



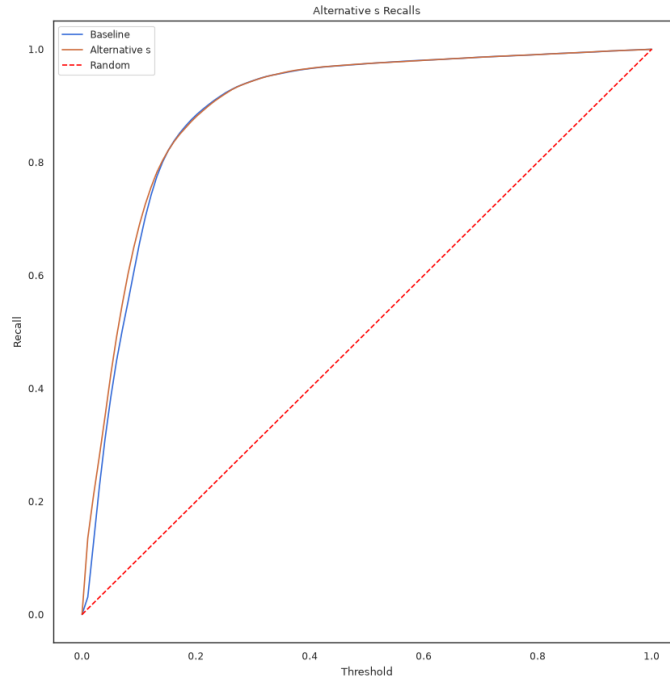
9 Multiplicative Score Decomposition

Problem: the baseline conflates two axes — explore/exploit and historic/current data — into one linear term.

$$s_j(t) = \pi_j(t) u_j(t)$$

$$\pi_j(t) = \alpha(t)\phi_j + (1-\alpha(t))f_j(t), \quad u_j(t) = \beta(t)p_j + (1-\beta(t)) \cdot 1$$

Result: R@5/15/25/35% = 0.425 / 0.821 / 0.922 / 0.959 AUC = 0.900



10 Binary-to-TTO Model ($X \rightarrow Y$)

Problem: two recipients who both eventually open are treated identically by the SAE, regardless of how fast they did it — discarding available time-to-open signal.

$$Y_{i,j} = 2^{-\Delta_{i,j}/h_\delta} \in [0, 1], \quad \Delta_{i,j} = \text{observed send-to-open time}$$
$$\min_{E,D} \ell(Y, \sigma(XB_{E,D}))$$

Input space stays binary (x); only the reconstruction target is swapped for the TTO-decayed label Y . Since e_j is unchanged as a training-time input, p_j carries over **mechanically unchanged**; $f_j(t)$ and $s_j(t)$ require no further modification to run end-to-end.

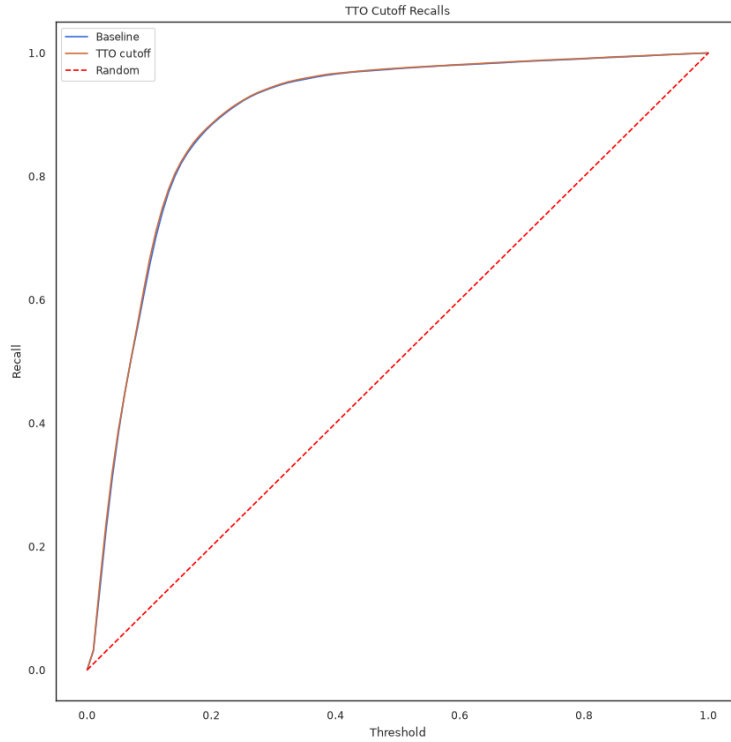
Status: proposed formulation, not yet benchmarked.

11 TTO Cutoff Thresholding

Problem: SAE trains on fully-converged opens, but active-learning queries $x(t)$ are always incomplete.

$$C_{i,j} = \mathbb{1}[\Delta_{i,j} \leq \delta_c], \quad \min_{E,D} \ell(X, \sigma(CB_{E,D}))$$

Result ($\delta_c = 720$): R@5/15/25/35% = 0.390 / 0.824 / 0.924 / 0.960 AUC = 0.895

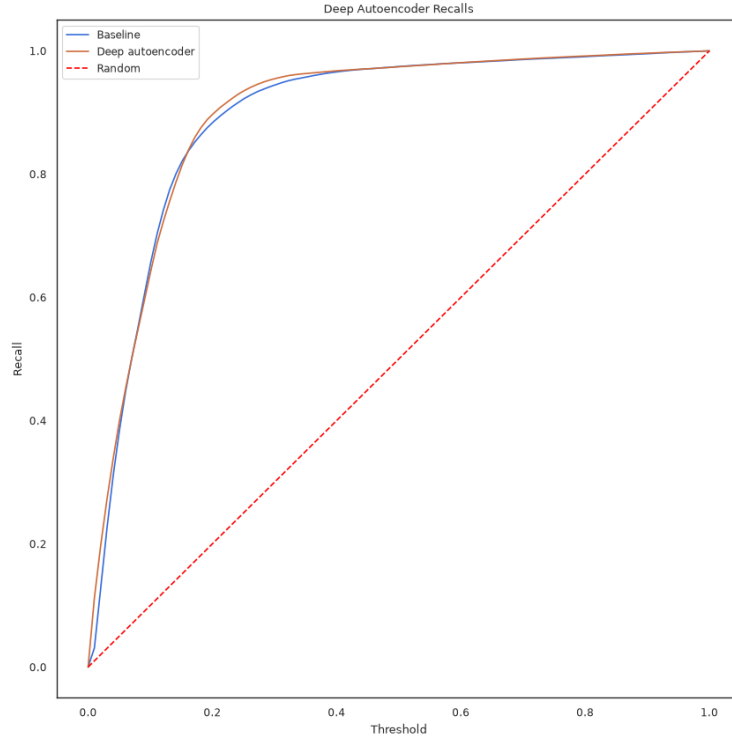


12 Nonlinear (Deep) Autoencoder

Problem: the linear SAE can only capture linear recipient–recipient relationships.

$$\text{Enc: } m \rightarrow 2d \rightarrow d, \quad \text{Dec: } d \rightarrow 2d \rightarrow m \quad (\text{ReLU, denoising mask})$$

Result ($d = 16$): $\text{R@5/15/25/35\%} = 0.400 / 0.815 / 0.936 / 0.963$ $\text{AUC} = 0.898$



13 Further Refinements (Untested)

- **Deep autoencoder regularization:** stronger latent-space dropout; dropout inside the encoder/decoder layers themselves, beyond the current denoising mask
- **Further TTO extensions:** a $Y \rightarrow Y$ model (continuous signal on the input side too, not just the target); $z(t)$ interpolation — a partially TTO-informed query state at inference instead of binary $x(t)$
- **Alternative p_j definitions:** clustering-based influence scores; time-dependent $p_j(t)$
- **Plug-and-play $s_j(t)$:** the multiplicative decomposition reuses the original primitives, so many of the above can be swapped in directly
- **TTO-based noise injection:** corrupt training rows using TTO-derived noise

14 Results Summary

Model	R@5%	R@15%	R@25%	R@35%	AUC
Baseline	0.385	0.821	0.923	0.958	0.894
Linear α schedule	0.383	0.823	0.927	0.960	0.894
Geometric α schedule	0.377	0.823	0.927	0.959	0.894
Confidence α schedule	0.341	0.818	0.926	0.960	0.891
Template weights	0.447	0.837	0.928	0.960	0.903
Forward-pass f	0.375	0.841	0.937	0.966	0.901
Variance-based p	0.479	0.837	0.926	0.960	0.905
Alternative s decomposition	0.425	0.821	0.922	0.959	0.900
TTO cutoff thresholding	0.390	0.824	0.924	0.960	0.895
Deep autoencoder	0.400	0.815	0.936	0.963	0.898

Standard deviations omitted for space; see whitepaper for full statistics.